

# AUTOM-07: CI/CD Integration

---

**Series:** AUTOM — Dynatrace Automation | **Notebook:** 7 of 8 | **Created:** January 2026 |  
**Last Updated:** 04/25/2026

CI/CD integration brings software development practices to Dynatrace configuration management. By storing configs in Git and deploying via pipelines, teams gain version control, review processes, and automated deployments.

---

## Table of Contents

---

1. [Introduction](#)
  2. [GitOps Fundamentals](#)
  3. [GitHub Actions](#)
    - [Terraform with Combined Auth](#)
    - [Vault Integration](#)
    - [Policy-as-Code Gates](#)
    - [Drift Detection](#)
    - [Reusable Workflows](#)
  4. [GitLab CI/CD](#)
  5. [ArgoCD Integration](#)
  6. [FluxCD Integration](#)
  7. [Dynatrace Operator GitOps Patterns](#)
  8. [Best Practices](#)
  9. [Governance Architecture](#)
  10. [Next Steps](#)
- 

## Prerequisites

---

Before starting this notebook, ensure you have:

| Requirement         | Description                                                                        |
|---------------------|------------------------------------------------------------------------------------|
| CI/CD Platform      | GitHub Actions, GitLab CI, or Jenkins                                              |
| Monaco or Terraform | One of the config-as-code tools                                                    |
| Git Repository      | For storing configurations                                                         |
| Authentication      | API Token + Platform Token for full coverage (see AUTOM-04)                        |
| HashiCorp Vault     | <i>Optional</i> — for runtime credential retrieval instead of static CI/CD secrets |
| OPA / Conftest      | <i>Optional</i> — for policy-as-code gates in pipelines                            |

## Token Types Reminder

As of Dynatrace Terraform provider **v1.88.0**, synthetic monitors and SLOs require a classic **API Token** ( `dt0c01` ). For full resource coverage in pipelines, use **Platform Token + API Token** together. See AUTOM-04: Terraform Provider for details.

---

## Learning Objectives

By the end of this notebook, you will:

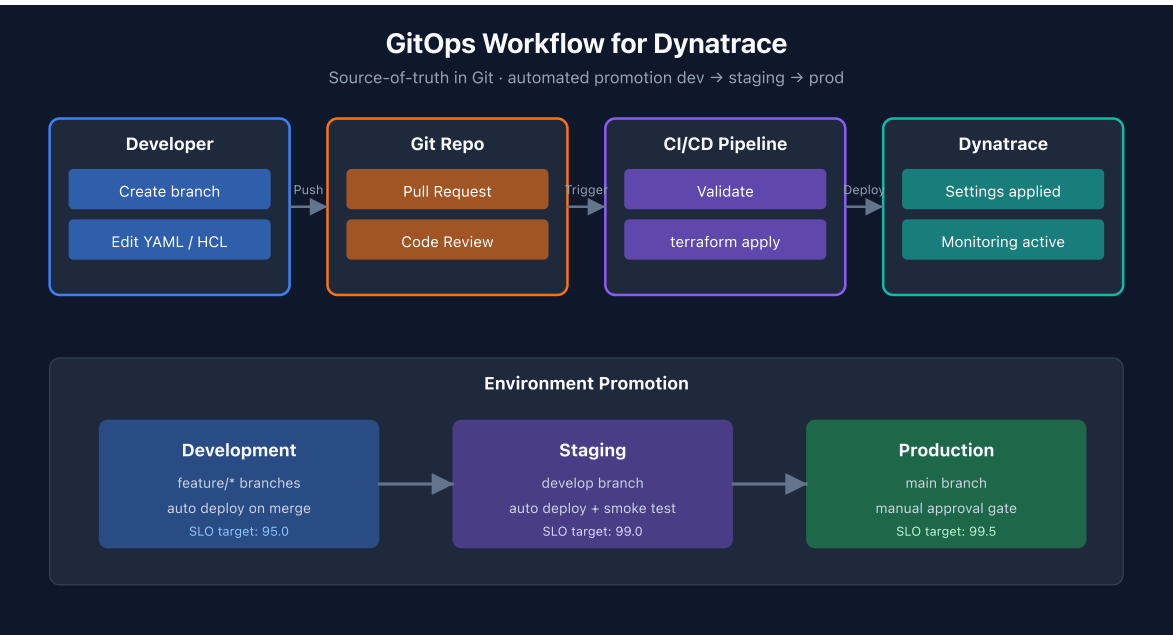
- Understand GitOps principles for Dynatrace
  - Know how to set up CI/CD pipelines for config deployment
  - Be able to implement pull request workflows with plan-as-PR-comment
  - Handle multi-environment deployments with combined auth
  - Integrate HashiCorp Vault for runtime credential retrieval
  - Implement policy-as-code gates with OPA/Conftest and Sentinel
  - Set up automated drift detection workflows
  - Create reusable GitHub Actions workflows for multi-team organizations
-

# 1. Introduction

## Why CI/CD for Dynatrace?

| Benefit         | Description                        |
|-----------------|------------------------------------|
| Version Control | All changes tracked in Git history |
| Code Review     | Pull requests for config changes   |
| Automation      | No manual deployments              |
| Rollback        | Revert to any previous state       |
| Audit Trail     | Who changed what, when             |

## GitOps Workflow



## 2. GitOps Fundamentals

### Repository Structure

```
dynatrace-config/
├── .github/
│   └── workflows/
│       ├── validate.yml
│       └── deploy.yml
├── manifest.yaml           # Monaco manifest
├── environments/
│   ├── dev.yaml
│   ├── staging.yaml
│   └── production.yaml
├── projects/
│   ├── alerting/
│   ├── management-zones/
│   └── synthetic/
└── README.md
```

### Branch Strategy

| Branch    | Purpose            | Deploys To     |
|-----------|--------------------|----------------|
| main      | Production configs | Production     |
| develop   | Integration branch | Staging        |
| feature/* | New features       | Dev (optional) |

### Environment Promotion



## 3. GitHub Actions

---

### Validation Workflow

**.github/workflows/validate.yml:**

```
name: Validate Dynatrace Config

on:
  pull_request:
    branches: [main, develop]
    paths:
      - 'projects/**'
      - 'manifest.yaml'

jobs:
  validate:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Install Monaco
        run: |
          curl -L https://github.com/Dynatrace/dynatrace-configuration-as-code/releases/latest/download/monaco-linux-amd64 -o monaco
          chmod +x monaco
          sudo mv monaco /usr/local/bin/

      - name: Validate Configuration
        run: monaco validate manifest.yaml

      - name: Dry Run Deploy
        env:
          DT_DEV_URL: ${ secrets.DT_DEV_URL }
          DT_DEV_TOKEN: ${ secrets.DT_DEV_TOKEN }
        run: |
          monaco deploy manifest.yaml --environment development --dry-run
```

### Deployment Workflow

**.github/workflows/deploy.yml:**

```
name: Deploy Dynatrace Config

on:
  push:
    branches: [main]
    paths:
      - 'projects/**'
      - 'manifest.yaml'

jobs:
  deploy-staging:
    runs-on: ubuntu-latest
    environment: staging
    steps:
      - uses: actions/checkout@v4

      - name: Install Monaco
        run: |
          curl -L https://github.com/Dynatrace/dynatrace-configuration-as-code/releases/latest/download/monaco-linux-amd64 -o monaco
          chmod +x monaco
          sudo mv monaco /usr/local/bin/

      - name: Deploy to Staging
        env:
          DT_STAGING_URL: ${ secrets.DT_STAGING_URL }
          DT_STAGING_TOKEN: ${ secrets.DT_STAGING_TOKEN }
        run: |
          monaco deploy manifest.yaml --environment staging

  deploy-production:
    needs: deploy-staging
    runs-on: ubuntu-latest
    environment: production
    steps:
      - uses: actions/checkout@v4

      - name: Install Monaco
        run: |
          curl -L https://github.com/Dynatrace/dynatrace-configuration-as-code/releases/latest/download/monaco-linux-amd64 -o monaco
          chmod +x monaco
          sudo mv monaco /usr/local/bin/

      - name: Deploy to Production
        env:
          DT_PROD_URL: ${ secrets.DT_PROD_URL }
          DT_PROD_TOKEN: ${ secrets.DT_PROD_TOKEN }
        run: |
          monaco deploy manifest.yaml --environment production
```

---

## Terraform Workflow with Combined Auth

For Terraform-based deployments, use **Platform Token + API Token** together for full resource coverage. The provider automatically routes each resource to the correct token.

```
name: Terraform Dynatrace

on:
  pull_request:
    branches: [main]
  push:
    branches: [main]

jobs:
  terraform:
    runs-on: ubuntu-latest
    permissions:
      contents: read
      pull-requests: write
    steps:
      - uses: actions/checkout@v4

      - name: Setup Terraform
        uses: hashicorp/setup-terraform@v3
        with:
          terraform_version: 1.6.0

      - name: Terraform Init
        run: terraform init

      - name: Terraform Plan
        id: plan
        if: github.event_name == 'pull_request'
        run: terraform plan -no-color -out=tfplan
        env:
          DYNATRACE_ENV_URL: ${ secrets.DT_ENV_URL }
          DYNATRACE_PLATFORM_TOKEN: ${ secrets.DT_PLATFORM_TOKEN }
          DYNATRACE_API_TOKEN: ${ secrets.DT_API_TOKEN }
          DYNATRACE_HTTP_OAUTH_PREFERENCE: "true"

      - name: Post Plan to PR
        if: github.event_name == 'pull_request'
        uses: borcherio/terraform-plan-comment@v2
        with:
          token: ${ github.token }
          planfile: tfplan

      - name: Terraform Apply
        if: github.ref == 'refs/heads/main' && github.event_name == 'push'
        run: terraform apply -auto-approve
        env:
          DYNATRACE_ENV_URL: ${ secrets.DT_ENV_URL }
          DYNATRACE_PLATFORM_TOKEN: ${ secrets.DT_PLATFORM_TOKEN }
          DYNATRACE_API_TOKEN: ${ secrets.DT_API_TOKEN }
          DYNATRACE_HTTP_OAUTH_PREFERENCE: "true"
```



| Secret            | Token Type  | Covers                                                        |
|-------------------|-------------|---------------------------------------------------------------|
| DT_PLATFORM_TOKEN | dt0s16.xxxx | Settings 2.0 + Gen3 Platform (workflows, documents, segments) |
| DT_API_TOKEN      | dt0c01.xxxx | Synthetic monitors, SLOs (v1.88.0 requirement)                |
| DT_ENV_URL        | —           | Tenant URL                                                    |

**Why combined auth?** A single API Token cannot manage Gen3 resources. A single Platform Token cannot manage synthetics/SLOs (removed in v1.88.0). Using both together gives the pipeline full coverage.

## Vault Integration for Runtime Credentials

For enterprise environments with **short-lived tokens** (e.g., 24-hour expiration), retrieve credentials at runtime from **HashiCorp Vault** instead of storing them as static CI/CD secrets.

```

name: Terraform with Vault Credentials

on:
  pull_request:
    branches: [main]
  push:
    branches: [main]

jobs:
  terraform:
    runs-on: ubuntu-latest
    permissions:
      id-token: write      # Required for OIDC → Vault JWT auth
      contents: read
      pull-requests: write
    steps:
      - uses: actions/checkout@v4

      - name: Retrieve Dynatrace credentials from Vault
        uses: hashicorp/vault-action@v3
        with:
          url: ${ secrets.VAULT_ADDR }
          method: jwt
          role: dynatrace-deployer
          secrets: |
            secret/data/dynatrace/prod platform_token | DYNATRACE_PLATFORM_TOKEN ;
            secret/data/dynatrace/prod api_token | DYNATRACE_API_TOKEN ;
            secret/data/dynatrace/prod env_url | DYNATRACE_ENV_URL

      - name: Setup Terraform
        uses: hashicorp/setup-terraform@v3

      - name: Terraform Init
        run: terraform init

      - name: Terraform Plan
        run: terraform plan -no-color -out=tfplan
        env:
          DYNATRACE_HTTP_OAUTH_PREFERENCE: "true"

      - name: Terraform Apply
        if: github.ref == 'refs/heads/main' && github.event_name == 'push'
        run: terraform apply -auto-approve
        env:
          DYNATRACE_HTTP_OAUTH_PREFERENCE: "true"

```

**Vault path structure per team per tenant:**

```

secret/
└─ dynatrace/
  │ └─ dev/
  │   │ └─ platform_token    # dt0s16 for dev tenant
  │   │ └─ api_token        # dt0c01 for dev tenant
  │   └─ env_url            # https://{dev-tenant}.live.dynatrace.com
  │ └─ prod/
  │   │ └─ platform_token
  │   │ └─ api_token
  │   └─ env_url
  └─ teams/
      │ └─ payments/        # Team-specific OAuth clients
      │   │ └─ client_id
      │   └─ client_secret
      └─ platform/
          │ └─ client_id
          └─ client_secret

```

**Why Vault?** Static GitHub Secrets don't expire. If your organization uses 24-hour token rotation or needs audit trails for credential access, Vault provides runtime retrieval with automatic expiration and access logging.

## Policy-as-Code Gates (OPA / Sentinel)

Add governance checks to your pipeline using **OPA/Conftest** (open-source) or **Sentinel** (Terraform Enterprise/Cloud).

### OPA/Conftest — Resource Type Allowlist

Restrict which Dynatrace resource types a team can create:

**policy/allowed\_resources.rego:**

```

package main

# Only allow these Dynatrace resource types
allowed_resources := {
  "dynatrace_alerting",
  "dynatrace_autotag_v2",
  "dynatrace_management_zone_v2",
  "dynatrace_maintenance"
}

deny[msg] {
  resource := input.planned_values.root_module.resources[_]
  startswith(resource.type, "dynatrace_")
  not allowed_resources[resource.type]
  msg := sprintf("Resource type '%s' is not in the allowlist", [resource.type])
}

```

## OPA/Conftest — Mandatory Team Tagging

Ensure all resources include team ownership metadata:

**policy/mandatory\_tags.rego:**

```

package main

deny[msg] {
  resource := input.planned_values.root_module.resources[_]
  resource.type == "dynatrace_autotag_v2"
  not resource.values.name
  msg := "Auto-tag resources must have a name"
}

deny[msg] {
  resource := input.planned_values.root_module.resources[_]
  startswith(resource.type, "dynatrace_")
  not has_team_ownership(resource)
  msg := sprintf("Resource '%s' must include team ownership metadata",
    [resource.address])
}

has_team_ownership(resource) {
  contains(resource.values.name, resource.values.team_name)
}

```

## Pipeline Integration

```
# Add after terraform plan step
- name: Install Conftest
  run: |
    wget -q https://github.com/open-policy-agent/conftest/releases/latest/download/conftest_Linux_x86_64.tar.gz
    tar xzf conftest_Linux_x86_64.tar.gz
    sudo mv conftest /usr/local/bin/

- name: Run Policy Checks
  run: |
    terraform show -json tfplan > tfplan.json
    conftest test tfplan.json --policy policy/
```

## Sentinel (Terraform Enterprise / Cloud)

For teams using **TFE** (now HCP Terraform), Sentinel policies enforce governance at the workspace level:

```
# sentinel/restrict_resource_types.sentinel
import "tfplan/v2" as tfplan

allowed_types = [
  "dynatrace_alerting",
  "dynatrace_autotag_v2",
  "dynatrace_management_zone_v2",
]

main = rule {
  all tfplan.resource_changes as _, rc {
    rc.type in allowed_types or not (rc.type matches "dynatrace_.*")
  }
}
```

**Important — Sentinel Limitations:** Sentinel evaluates Terraform **plans** — it does not grant or restrict Dynatrace API access at runtime. It cannot reduce the permissions of a token that has broad scope. If a pipeline token can write all Synthetic monitors, Sentinel cannot narrow that. Use **Dynatrace IAM policies** (see **AUTOM-04: IAM Policy Management**) for platform-level access control, and Sentinel for pipeline-level governance. Sentinel only runs in **HCP Terraform** — it is not available in open-source Terraform, GitHub Actions, or plain CI runners. For non-HCP environments, use **OPA/Conftest** as shown above.

## Drift Detection Workflow

Schedule automatic drift detection to catch manual UI changes ("ClickOps") that bypass your Terraform pipeline:

```

name: Drift Detection

on:
  schedule:
    - cron: '0 6 * * 1-5' # Every weekday at 6am UTC
  workflow_dispatch: {} # Allow manual trigger

jobs:
  detect-drift:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Setup Terraform
        uses: hashicorp/setup-terraform@v3

      - name: Terraform Init
        run: terraform init

      - name: Check for Drift
        id: drift
        run: |
          set -e
          terraform plan -detailed-exitcode -no-color 2>&1 | tee drift-
output.txt
          echo "exitcode=$?" >&& "$GITHUB_OUTPUT"
        env:
          DYNATRACE_ENV_URL: ${ secrets.DT_ENV_URL }
          DYNATRACE_PLATFORM_TOKEN: ${ secrets.DT_PLATFORM_TOKEN }
          DYNATRACE_API_TOKEN: ${ secrets.DT_API_TOKEN }
          DYNATRACE_HTTP_OAUTH_PREFERENCE: "true"

      - name: Create Issue on Drift
        if: steps.drift.outputs.exitcode == '2'
        uses: actions/github-script@v7
        with:
          script: |
            const fs = require('fs');
            const drift = fs.readFileSync('drift-output.txt', 'utf8').slice(-3000);
            await github.rest.issues.create({
              owner: context.repo.owner,
              repo: context.repo.repo,
              title: '⚠️ Dynatrace configuration drift detected',
              body: `Scheduled drift detection found changes not in Terraform
state.\n\n\`\`\`\n${drift}\n\n\`\`\`\n\nSomeone may have made manual changes in the
Dynatrace UI.` ,
              labels: ['drift', 'dynatrace']
            });

```

**Exit codes:** `terraform plan -detailed-exitcode` returns `0` = no changes, `1` = error, `2` = drift detected. This lets your pipeline take different actions based on the result.

---

## Reusable GitHub Actions Workflows

For organizations managing multiple Dynatrace tenants or LOB repositories, create **reusable workflows** that any repo can call:

**.github/workflows/terraform-dynatrace.yml** (in a shared workflow repo):



```

name: Terraform Dynatrace (Reusable)

on:
  workflow_call:
    inputs:
      working_directory:
        required: false
        type: string
        default: '.'
      terraform_version:
        required: false
        type: string
        default: '1.6.0'
    secrets:
      DT_ENV_URL:
        required: true
      DT_PLATFORM_TOKEN:
        required: true
      DT_API_TOKEN:
        required: true

jobs:
  plan:
    runs-on: ubuntu-latest
    permissions:
      contents: read
      pull-requests: write
    defaults:
      run:
        working-directory: ${ inputs.working_directory }
    steps:
      - uses: actions/checkout@v4

      - name: Setup Terraform
        uses: hashicorp/setup-terraform@v3
        with:
          terraform_version: ${ inputs.terraform_version }

      - name: Terraform Init
        run: terraform init

      - name: Terraform Plan
        run: terraform plan -no-color -out=tfplan
        env:
          DYNATRACE_ENV_URL: ${ secrets.DT_ENV_URL }
          DYNATRACE_PLATFORM_TOKEN: ${ secrets.DT_PLATFORM_TOKEN }
          DYNATRACE_API_TOKEN: ${ secrets.DT_API_TOKEN }
          DYNATRACE_HTTP_OAUTH_PREFERENCE: "true"

      - name: Post Plan to PR
        if: github.event_name == 'pull_request'

```

```

    uses: borchero/terraform-plan-comment@v2
    with:
      token: ${ github.token }
      planfile: tfplan

  apply:
    needs: plan
    if: github.ref == 'refs/heads/main' && github.event_name == 'push'
    runs-on: ubuntu-latest
    defaults:
      run:
        working-directory: ${ inputs.working_directory }
    steps:
      - uses: actions/checkout@v4

      - name: Setup Terraform
        uses: hashicorp/setup-terraform@v3
        with:
          terraform_version: ${ inputs.terraform_version }

      - name: Terraform Init
        run: terraform init

      - name: Terraform Apply
        run: terraform apply -auto-approve
        env:
          DYNATRACE_ENV_URL: ${ secrets.DT_ENV_URL }
          DYNATRACE_PLATFORM_TOKEN: ${ secrets.DT_PLATFORM_TOKEN }
          DYNATRACE_API_TOKEN: ${ secrets.DT_API_TOKEN }
          DYNATRACE_HTTP_OAUTH_PREFERENCE: "true"

```

**Calling the reusable workflow** from a LOB repo:

```
# .github/workflows/deploy.yml (in dt-lob-payments repo)
name: Deploy Dynatrace Config

on:
  pull_request:
    branches: [main]
  push:
    branches: [main]

jobs:
  terraform:
    uses: my-org/dt-templates/.github/workflows/terraform-dynatrace.yml@main
    with:
      working_directory: terraform/
    secrets:
      DT_ENV_URL: ${ secrets.DT_ENV_URL }
      DT_PLATFORM_TOKEN: ${ secrets.DT_PLATFORM_TOKEN }
      DT_API_TOKEN: ${ secrets.DT_API_TOKEN }
```

**Multi-tenant pattern:** For organizations with 5+ tenants, the reusable workflow is called once per environment with different secrets, providing a consistent deployment experience across all tenants.

---

## 4. GitLab CI/CD

---

### Pipeline Configuration

**.gitlab-ci.yml:**

```

stages:
  - validate
  - deploy-staging
  - deploy-production

variables:
  MONACO_VERSION: "2.0.0"

.monaco-setup: &monaco-setup
  before_script:
    - curl -L https://github.com/Dynatrace/dynatrace-configuration-as-code/releases/download/v${MONACO_VERSION}/monaco-linux-amd64 -o monaco
    - chmod +x monaco
    - mv monaco /usr/local/bin/

validate:
  stage: validate
  <<: *monaco-setup
  script:
    - monaco validate manifest.yaml
    - monaco deploy manifest.yaml --environment development --dry-run
  rules:
    - if: $CI_PIPELINE_SOURCE == "merge_request_event"

deploy-staging:
  stage: deploy-staging
  <<: *monaco-setup
  script:
    - monaco deploy manifest.yaml --environment staging
  environment:
    name: staging
  rules:
    - if: $CI_COMMIT_BRANCH == "main"

deploy-production:
  stage: deploy-production
  <<: *monaco-setup
  script:
    - monaco deploy manifest.yaml --environment production
  environment:
    name: production
  rules:
    - if: $CI_COMMIT_BRANCH == "main"
      when: manual

```

## Variables Configuration

In GitLab: **Settings** → **CI/CD** → **Variables**

| Variable         | Protected | Masked |
|------------------|-----------|--------|
| DT_DEV_URL       | No        | No     |
| DT_DEV_TOKEN     | No        | Yes    |
| DT_STAGING_URL   | No        | No     |
| DT_STAGING_TOKEN | Yes       | Yes    |
| DT_PROD_URL      | Yes       | No     |
| DT_PROD_TOKEN    | Yes       | Yes    |

---

## 5. ArgoCD Integration

---

For Kubernetes-native GitOps, use ArgoCD with Dynatrace configs.

### ArgoCD Application for Monaco Configs

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: dynatrace-config
  namespace: argocd
spec:
  project: default
  source:
    repoURL: https://github.com/org/dynatrace-config.git
    targetRevision: HEAD
    path: projects
  destination:
    server: https://kubernetes.default.svc
    namespace: dynatrace
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
```

### ArgoCD for Dynatrace Operator (DynaKube)

Deploy and manage the Dynatrace Operator via ArgoCD:

```

apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: dynatrace-operator
  namespace: argocd
spec:
  project: default
  source:
    repoURL: https://github.com/org/k8s-platform.git
    targetRevision: HEAD
    path: dynatrace
  destination:
    server: https://kubernetes.default.svc
    namespace: dynatrace
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
    syncOptions:
      - CreateNamespace=true

```

### Repository structure for Dynatrace Operator:

```

dynatrace/
├─ kustomization.yaml
├─ namespace.yaml
├─ operator.yaml          # Dynatrace Operator deployment
├─ dynakube.yaml          # DynaKube custom resource
├─ secrets/
└─┬─ dynakube-secret.yaml # External Secrets reference

```

## External Secrets for Token Management

Use External Secrets Operator to sync tokens from secret stores:

```

apiVersion: external-secrets.io/v1beta1
kind: ExternalSecret
metadata:
  name: dynakube-tokens
  namespace: dynatrace
spec:
  refreshInterval: 1h
  secretStoreRef:
    name: aws-secrets-manager
    kind: ClusterSecretStore
  target:
    name: dynakube
    creationPolicy: Owner
  data:
    - secretKey: apiToken
      remoteRef:
        key: dynatrace/api-token
    - secretKey: dataIngestToken
      remoteRef:
        key: dynatrace/data-ingest-token

```

## Using Config Management Plugins

**argocd-cm ConfigMap:**

```

apiVersion: v1
kind: ConfigMap
metadata:
  name: argocd-cm
  namespace: argocd
data:
  configManagementPlugins: |
    - name: monaco
      generate:
        command: ["sh", "-c"]
        args: ["monaco deploy manifest.yaml --environment $ARGOCD_ENV_ENVIRONMENT"]

```

---

## 6. FluxCD Integration

---

FluxCD provides an alternative GitOps approach with a pull-based reconciliation model.

## FluxCD vs ArgoCD

| Feature          | FluxCD                 | ArgoCD                         |
|------------------|------------------------|--------------------------------|
| Architecture     | Controller-based       | Server + UI                    |
| UI               | Minimal (Weave GitOps) | Built-in web UI                |
| Multi-tenancy    | Namespace-based        | Project-based                  |
| Helm Support     | Native HelmRelease     | Application CRD                |
| Image Automation | Built-in               | Requires Argo CD Image Updater |

## FluxCD for Dynatrace Operator

**GitRepository source:**

```
apiVersion: source.toolkit.fluxcd.io/v1
kind: GitRepository
metadata:
  name: dynatrace-config
  namespace: flux-system
spec:
  interval: 1m
  url: https://github.com/org/k8s-platform
  ref:
    branch: main
  secretRef:
    name: github-token
```

**Kustomization for Dynatrace:**



```
apiVersion: kustomize.toolkit.fluxcd.io/v1
kind: Kustomization
metadata:
  name: dynatrace
  namespace: flux-system
spec:
  interval: 10m
  targetNamespace: dynatrace
  sourceRef:
    kind: GitRepository
    name: dynatrace-config
  path: ./dynatrace
  prune: true
  healthChecks:
    - apiVersion: apps/v1
      kind: Deployment
      name: dynatrace-operator
      namespace: dynatrace
```

## FluxCD HelmRelease for Dynatrace Operator

Deploy via Helm with FluxCD:

```
apiVersion: source.toolkit.fluxcd.io/v1beta2
kind: HelmRepository
metadata:
  name: dynatrace
  namespace: flux-system
spec:
  interval: 1h
  url: https://raw.githubusercontent.com/Dynatrace/dynatrace-
operator/main/config/helm/repos/stable

---
apiVersion: helm.toolkit.fluxcd.io/v2beta1
kind: HelmRelease
metadata:
  name: dynatrace-operator
  namespace: dynatrace
spec:
  interval: 5m
  chart:
    spec:
      chart: dynatrace-operator
      version: ">=1.0.0"
      sourceRef:
        kind: HelmRepository
        name: dynatrace
        namespace: flux-system
  values:
    installCRD: true
```

## SOPS for Secret Management with FluxCD

Use SOPS to encrypt secrets in Git:

```
apiVersion: kustomize.toolkit.fluxcd.io/v1
kind: Kustomization
metadata:
  name: dynatrace-secrets
  namespace: flux-system
spec:
  interval: 10m
  sourceRef:
    kind: GitRepository
    name: dynatrace-config
  path: ./dynatrace/secrets
  prune: true
  decryption:
    provider: sops
    secretRef:
      name: sops-age
```

---

## 7. Dynatrace Operator GitOps Patterns

---

When deploying the Dynatrace Operator via GitOps, follow these patterns for production environments.

**Important:** Use `apiVersion: dynatrace.com/v1beta5` or `v1beta6` for Dynatrace Operator 1.8.x. Earlier versions (v1beta1, v1beta2) are deprecated and no longer supported.

### Multi-Cluster Deployment Pattern

For organizations with multiple Kubernetes clusters:

```

platform-gitops/
├─ clusters/
│   ├── production-east/
│   │   └─ dynatrace/
│   │       ├── kustomization.yaml
│   │       └─ dynakube-patch.yaml    # Cluster-specific overrides
│   ├── production-west/
│   │   └─ dynatrace/
│   │       ├── kustomization.yaml
│   │       └─ dynakube-patch.yaml
│   └─ staging/
│       └─ dynatrace/
│           ├── kustomization.yaml
│           └─ dynakube-patch.yaml
└─ base/
    └─ dynatrace/
        ├── kustomization.yaml
        ├── namespace.yaml
        ├── operator.yaml
        └─ dynakube.yaml              # Base DynaKube config

```

#### Base kustomization.yaml:

```

apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources:
  - namespace.yaml
  - https://github.com/Dynatrace/dynatrace-
operator/releases/download/v1.8.1/kubernetes.yaml
  - dynakube.yaml

```

#### Cluster overlay (production-east):

```

apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources:
  - ../../base/dynatrace
patchesStrategicMerge:
  - dynakube-patch.yaml

```

#### Cluster-specific patch:

```

apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  oneAgent:
    cloudNativeFullStack:
      args:
        - --set-host-group=production-east

```

## Multi-Tenant Observability Pattern

For clusters serving multiple teams with different Dynatrace tenants:

```

# Team A - uses tenant-a.live.dynatrace.com
apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: team-a-dynakube
  namespace: dynatrace
spec:
  apiUrl: https://tenant-a.live.dynatrace.com/api
  tokens: team-a-tokens
  oneAgent:
    cloudNativeFullStack:
      namespaceSelector:
        matchLabels:
          dynatrace-tenant: team-a

---
# Team B - uses tenant-b.live.dynatrace.com
apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: team-b-dynakube
  namespace: dynatrace
spec:
  apiUrl: https://tenant-b.live.dynatrace.com/api
  tokens: team-b-tokens
  oneAgent:
    cloudNativeFullStack:
      namespaceSelector:
        matchLabels:
          dynatrace-tenant: team-b

```

## GitOps Health Checks

Configure sync health checks to verify Dynatrace deployment:

### ArgoCD health check:

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: dynatrace
spec:
  # ... source config ...
  ignoreDifferences:
    - group: dynatrace.com
      kind: DynaKube
      jsonPointers:
        - /status # Ignore status field changes
```

### FluxCD health check:

```
apiVersion: kustomize.toolkit.fluxcd.io/v1
kind: Kustomization
metadata:
  name: dynatrace
spec:
  healthChecks:
    - apiVersion: apps/v1
      kind: Deployment
      name: dynatrace-operator
      namespace: dynatrace
    - apiVersion: apps/v1
      kind: DaemonSet
      name: dynakube-oneagent
      namespace: dynatrace
```

## Sealed Secrets Alternative

For Kubernetes-native secret encryption:

```
apiVersion: bitnami.com/v1alpha1
kind: SealedSecret
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  encryptedData:
    apiToken: AgBy8BYo...encrypted...
    dataIngestToken: AgCtr4s2...encrypted...
```

Generate sealed secrets:

```
# Seal the secret (use your actual tokens from Dynatrace)
kubectl create secret generic dynakube \
  --from-literal=apiToken=<your-api-token> \
  --from-literal=dataIngestToken=<your-data-ingest-token> \
  --dry-run=client -o yaml | \
  kubeseal --format yaml > sealed-dynakube.yaml
```

---

## 8. Best Practices

---

### Security

| Practice                       | Description                                                                   |
|--------------------------------|-------------------------------------------------------------------------------|
| <b>Masked secrets</b>          | Always mask API tokens in CI/CD                                               |
| <b>Protected branches</b>      | Require reviews for main branch                                               |
| <b>Environment protection</b>  | Manual approval for production                                                |
| <b>Token rotation</b>          | Rotate tokens regularly                                                       |
| <b>Least privilege</b>         | Minimal token scopes                                                          |
| <b>External Secrets</b>        | Use ESO, SOPS, or Sealed Secrets for K8s                                      |
| <b>Vault for runtime creds</b> | Retrieve tokens at pipeline runtime instead of static CI/CD secrets           |
| <b>Combined auth</b>           | Use Platform Token + API Token for full resource coverage (v1.88.0)           |
| <b>Policy-as-code</b>          | OPA/Conftest or Sentinel to enforce resource allowlists and mandatory tagging |

| Practice                  | Description                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------|
| <b>State file access</b>  | Lock down HCP Terraform workspace permissions — state files may contain OAuth credentials |
| <b>Two-pipeline model</b> | Separate IAM management (central team) from config management (LOB teams)                 |
| <b>Single SA writer</b>   | Only the pipeline SA writes to production; all humans get read-only access                |

## Workflow Design

| Practice                  | Description                                                                  |
|---------------------------|------------------------------------------------------------------------------|
| <b>Validate first</b>     | Always validate before deploy                                                |
| <b>Dry run PRs</b>        | Show what would change                                                       |
| <b>Plan as PR comment</b> | Post <code>terraform plan</code> output directly on PRs for reviewer context |
| <b>Staged rollout</b>     | Dev → Staging → Production                                                   |
| <b>Manual gates</b>       | Require approval for production                                              |
| <b>Rollback plan</b>      | Know how to revert changes                                                   |
| <b>Health checks</b>      | Verify deployment success                                                    |
| <b>Drift detection</b>    | Schedule <code>terraform plan -detailed-exitcode</code> to catch ClickOps    |
| <b>Reusable workflows</b> | <code>workflow_call</code> for consistent deployment across repos            |

## GitOps Tool Selection

| Use Case                        | Recommended Tool           |
|---------------------------------|----------------------------|
| <b>Simple CI/CD</b>             | GitHub Actions / GitLab CI |
| <b>K8s-native, UI preferred</b> | ArgoCD                     |
| <b>K8s-native, Helm-first</b>   | FluxCD                     |
| <b>Multi-cluster enterprise</b> | ArgoCD or FluxCD           |

## Pull Request Template

[.github/PULL\\_REQUEST\\_TEMPLATE.md](#):



```
<a id="dynatrace-configuration-change"></a>
## Dynatrace Configuration Change
### Description
<!-- What configuration is being changed? -->

### Type of Change
- [ ] New configuration
- [ ] Modification to existing config
- [ ] Deletion of config

### Environments Affected
- [ ] Development
- [ ] Staging
- [ ] Production

### Validation
- [ ] Monaco validate passed
- [ ] Dry run successful
- [ ] Tested in dev environment

### Rollback Plan
<!-- How to revert if issues arise? -->
```

---

## Notification Integration

Add deployment notifications:

```
# GitHub Actions notification step
- name: Notify Slack
  if: always()
  uses: slackapi/slack-github-action@v1
  with:
    payload: |
      {
        "text": "Dynatrace config deployment: ${ job.status }",
        "blocks": [
          {
            "type": "section",
            "text": {
              "type": "mrkdown",
              "text": "*Deployment to ${ github.ref_name }*\nStatus: ${ job.status }\nCommit: ${ github.sha }"
            }
          }
        ]
      }
    env:
      SLACK_WEBHOOK_URL: ${ secrets.SLACK_WEBHOOK }
```

## 9. Governance Architecture

The previous sections covered individual governance tools (Vault, OPA, Sentinel, drift detection). This section synthesizes them into a cohesive **enterprise governance architecture** for Dynatrace configuration management.

### The Two-Pipeline Model

Enterprise Dynatrace governance requires **two distinct pipelines** with separate identities, scopes, and cadences:

#### Pipeline A — IAM Pipeline ("Who can do what")

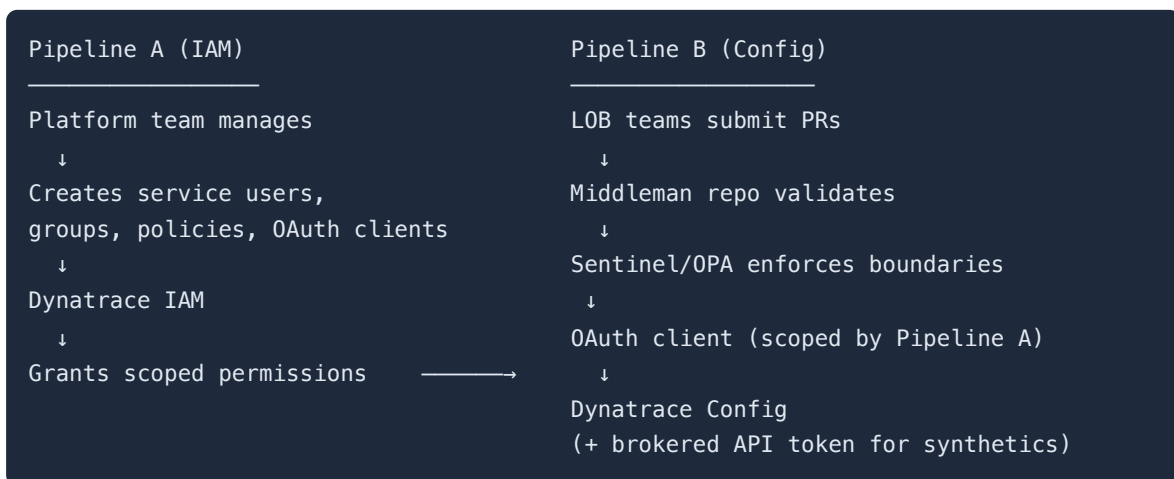
| Aspect          | Detail                                                                                               |
|-----------------|------------------------------------------------------------------------------------------------------|
| <b>Owner</b>    | Central Platform / Observability team                                                                |
| <b>Identity</b> | Service User + OAuth client scoped to IAM resources                                                  |
| <b>Manages</b>  | Groups, policies, policy bindings, environment-level permissions<br>( <code>dynatrace_iam_*</code> ) |

| Aspect            | Detail                                                                                              |
|-------------------|-----------------------------------------------------------------------------------------------------|
| <b>Governance</b> | Sentinel/OPA prevents privilege escalation (no <code>account:admin</code> , no cross-team bindings) |
| <b>Cadence</b>    | Infrequent — runs when teams onboard, roles change, or environments are provisioned                 |
| <b>Repo</b>       | <code>dt-global-config</code> or equivalent central IAM repo                                        |

## Pipeline B — Configuration Pipeline ("The actual work")

| Aspect            | Detail                                                                          |
|-------------------|---------------------------------------------------------------------------------|
| <b>Owner</b>      | Per-team / per-LOB (via middleman repo or brokered self-service)                |
| <b>Identity</b>   | Service User + OAuth client scoped to the config domains Pipeline A granted     |
| <b>Manages</b>    | Workflows, dashboards, segments, alerting profiles, auto-tags, management zones |
| <b>Auth</b>       | Dual-auth (OAuth + API token) when synthetics are in scope                      |
| <b>Governance</b> | Sentinel/OPA enforces naming, tagging, MZ boundaries, allowed resource types    |
| <b>Cadence</b>    | Frequent — runs on every config change promotion through environments           |
| <b>Repo</b>       | <code>dt-lob-&lt;lobname&gt;</code> or team-specific config repos               |

Pipeline B **can only do what Pipeline A has granted**. Its OAuth client's effective permissions are bounded by the IAM policies, group memberships, and environment scoping that Pipeline A defined.



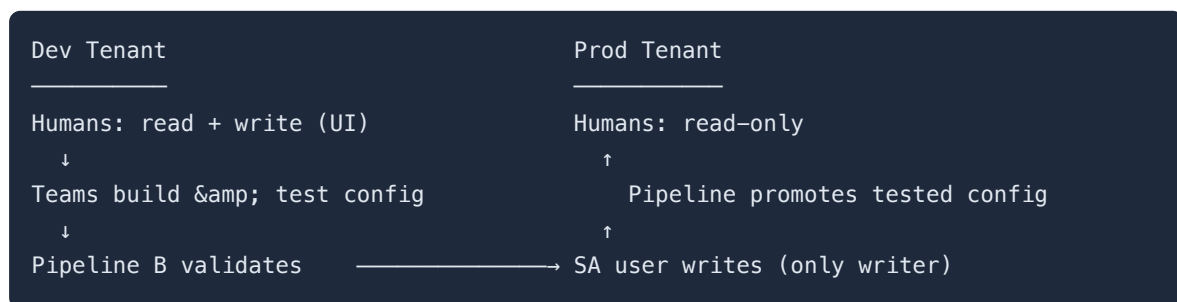
**Key principle:** Pipeline A creates the identities and permissions. Pipeline B operates under those grants. Separation of concerns between access governance and

configuration management.

## Operational Model: Single SA Writer

In practice, the two-pipeline architecture produces a clear operational rule: **all human users have read-only access in production, and only one service account can write.**

Teams build and validate configuration in a **dev/test tenant** where they have UI write access. Once tested, the provisioning pipeline — running as the single SA (service account) — promotes that configuration to production. No human ClickOps in prod, and a single auditable write path.



This model works with or without Sentinel. It directly addresses the access scoping challenge: within a single tenant, you cannot easily separate write access between teams for v1 resources. By splitting dev (where humans create) from prod (where only the pipeline writes), you get isolation through environment boundaries rather than token scoping. The SA's credentials live in Vault or HCP Terraform — never in human hands.

## Defense-in-Depth Governance Layers

No single tool provides complete governance. Enterprise Dynatrace configuration management uses **layered controls** where each layer compensates for the limitations of the others:

| Layer                          | Role                                                           | Tool / Mechanism                                      |
|--------------------------------|----------------------------------------------------------------|-------------------------------------------------------|
| <b>Dynatrace IAM</b>           | Source of truth for access control                             | IAM policies, groups, bindings (see <b>AUTOM-04</b> ) |
| <b>Service User + OAuth</b>    | Scoped pipeline identity for Gen3 resources                    | OAuth client credentials                              |
| <b>Dual Auth (OAuth + API)</b> | Full coverage when pipeline manages both Gen3 and v1 resources | Combined provider config                              |

| Layer                            | Role                                                    | Tool / Mechanism                                    |
|----------------------------------|---------------------------------------------------------|-----------------------------------------------------|
| <b>Sentinel / OPA</b>            | Governance guardrails on what Terraform can do          | Policy-as-code (see above)                          |
| <b>Vault</b>                     | Runtime credential management with expiration and audit | HashiCorp Vault (see above)                         |
| <b>PR Review + CODEOWNERS</b>    | Human approval gate                                     | GitHub/GitLab branch protection                     |
| <b>Module Allowlists</b>         | Only approved Terraform modules in use                  | Sentinel module restrictions or repo-level controls |
| <b>State File Access Control</b> | Prevent credential extraction from Terraform state      | HCP Terraform workspace permissions                 |

**State file risk:** If a team can read the state of a middleman workspace in HCP Terraform, they could potentially extract the OAuth client credentials or see IAM bindings they should not access. Lock down workspace-level permissions in HCP Terraform appropriately.

## What Sentinel / OPA Can and Cannot Do

| Sentinel / OPA CAN                                              | Sentinel / OPA CANNOT                                               |
|-----------------------------------------------------------------|---------------------------------------------------------------------|
| Restrict which Terraform modules and resource types may be used | Integrate with Dynatrace IAM at runtime                             |
| Enforce naming conventions, tagging, and MZ boundaries          | Reduce the runtime permissions of a Dynatrace token                 |
| Prevent privilege escalation in IAM-as-code                     | Replace Dynatrace RBAC / IAM policies                               |
| Validate team-to-object ownership in brokered self-service      | Run outside HCP Terraform (Sentinel only — use OPA/Conftest for CI) |

Sentinel does not *grant* access. Sentinel decides whether Terraform is *allowed to grant* access. Dynatrace IAM remains the source of truth for what the pipeline identity can actually do.

## Framing for Security Reviewers

When presenting this architecture to security teams or auditors, document the v1 Synthetic limitation as an **accepted platform constraint with compensating controls**:

1. **Gen3/platform resources** — genuinely scoped via Service User + OAuth + IAM policies
2. **v1 Synthetic monitors** — brokered API token access through central pipeline (teams never hold the token directly)
3. **Pipeline guardrails** — Sentinel/OPA enforce naming, tagging, MZ boundaries, and resource type restrictions
4. **Credential management** — Vault provides runtime retrieval, automatic expiration, and access audit logging
5. **Audit trail** — Git history + PR reviews + Dynatrace audit logs provide full change traceability

This framing is much stronger than trying to pretend the v1 API scoping gap does not exist. It demonstrates a mature, defense-in-depth approach to a known platform limitation.

---

## 10. Next Steps

---

### Deployment Event Tracking

Send deployment events to Dynatrace:

```
- name: Send Deployment Event
  run: |
    curl -X POST "${{ secrets.DT_URL }}/api/v2/events/ingest" \
      -H "Authorization: Api-Token ${{ secrets.DT_TOKEN }}" \
      -H "Content-Type: application/json" \
      -d '{
        "eventType": "CUSTOM_DEPLOYMENT",
        "title": "Dynatrace Config Deployment",
        "properties": {
          "source": "github-actions",
          "commit": "${{ github.sha }}",
          "branch": "${{ github.ref_name }}"
        }
      }'
```

## Continue the Series

| Next Notebook                         | Focus                                       |
|---------------------------------------|---------------------------------------------|
| <b>AUTOM-08: Migration Automation</b> | Bulk configuration transfer between tenants |

## Additional Resources

- [GitHub Actions Documentation](#)
- [GitLab CI/CD Documentation](#)
- [ArgoCD Documentation](#)
- [FluxCD Documentation](#)
- [Dynatrace Operator](#)
- [Monaco GitHub Repository](#)
- [External Secrets Operator](#)
- [OPA/Conftest](#)
- [Sentinel Documentation](#)
- [HashiCorp Vault Action](#)
- [Dynatrace OAuth Client Guide](#)

---

## Summary

In this notebook, you learned:

- GitOps fundamentals for Dynatrace configuration
- Setting up GitHub Actions and GitLab CI/CD pipelines
- **Combined auth** (Platform Token + API Token) for full resource coverage in Terraform pipelines
- **Vault integration** for runtime credential retrieval with short-lived tokens
- **Policy-as-code gates** using OPA/Conftest and Sentinel for governance enforcement
- **Sentinel limitations** — it governs Terraform plans, not Dynatrace API permissions at runtime
- **Drift detection** workflows to catch manual UI changes
- **Reusable workflows** ( `workflow_call` ) for multi-team organizations
- ArgoCD integration with External Secrets for token management
- FluxCD with HelmRelease and SOPS for secret encryption

- Dynatrace Operator GitOps patterns for multi-cluster and multi-tenant environments
- **Two-pipeline model:** Pipeline A (IAM) creates identities, Pipeline B (Config) applies configuration under those grants
- **Single SA writer:** only the pipeline SA writes to production; humans develop in dev/test
- **Defense-in-depth:** layered governance from Dynatrace IAM through Sentinel/OPA, Vault, PR gates, and state file controls

**Key Takeaway:** CI/CD integration transforms Dynatrace config management into a software development workflow. Use a two-pipeline model to separate IAM governance from configuration management. Enforce single-SA-writer in production. Layer Vault, OPA/Sentinel, and Dynatrace IAM policies for defense-in-depth — no single tool provides complete governance.

Continue to AUTOM-08: Migration Automation to learn bulk configuration transfer.

## Community Resources & CI/CD Examples

The following GitHub repositories provide working CI/CD pipelines, validation scripts, and platform engineering references:

### Pipeline Observability Samples (in `dynatrace-configuration-as-code-samples`)

Both Monaco and Terraform variants are available for each CI/CD platform:

| CI/CD Platform | Monaco Sample                              | Terraform Sample                                     |
|----------------|--------------------------------------------|------------------------------------------------------|
| GitHub Actions | <code>github_pipeline_observability</code> | <code>github_pipeline_observability_terraform</code> |
| GitLab CI      | <code>gitlab_pipeline_observability</code> | <code>gitlab_pipeline_observability_terraform</code> |
| Azure DevOps   | <code>azure_devops_observability</code>    | --                                                   |
| ArgoCD         | <code>argocd_observability</code>          | <code>argocd_observability_terraform</code>          |

All samples ingest SDLC events via OpenPipeline and include pre-built dashboards.



CI Validation Scripts (in `dynatrace-configuration-as-code-samples/scripts/ci/` )

| Script                             | Purpose                                |
|------------------------------------|----------------------------------------|
| <code>validate-monaco.sh</code>    | Validate Monaco configuration in CI    |
| <code>validate-terraform.sh</code> | Validate Terraform configuration in CI |
| <code>check-secrets.sh</code>      | Detect hardcoded secrets before commit |

CI/CD Automation Tools

| Repository                              | Description                                                                                      |
|-----------------------------------------|--------------------------------------------------------------------------------------------------|
| <code>dynatrace-automation-tools</code> | CLI for Site Reliability Guardian automation and deployment event ingestion in CI/CD (v1.0.3)    |
| <code>monaco-demo</code>                | Working GitHub Actions workflow for Monaco deploy with <code>.github/workflows/monaco.yml</code> |

Platform Engineering References

| Repository                             | Description                                                                                        |
|----------------------------------------|----------------------------------------------------------------------------------------------------|
| <code>platform-engineering-demo</code> | Full IDP reference: ArgoCD + Backstage + Keptn + OpenTelemetry + Dynatrace                         |
| <code>demo-crossplane</code>           | Crossplane + Terraform provider for continuous GitOps reconciliation (no manual pipeline triggers) |
| <code>obslab-composite-srg</code>      | Composite Site Reliability Guardians with GitHub Actions                                           |
| <code>obslab-release-validation</code> | Release validation using k6, business events, workflows, and SRG                                   |

*This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.*